

Literal Generic-Affine Möbius Dispersion: Exact TT^* Kernels, a Chowla-Level Uniformity Barrier, and the Restricted Arithmetic Hypothesis

Liang Wang
School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

We audit the first parity-facing arithmetic gate in TPC-MVP1. After the exact TPC-93 content extraction, a literal generic block has two primitive affine forms

$$D(z) = d + sz, \quad V(z) = u + az, \quad su - ad = h_0 \neq 0,$$

the extracted sign and coprimality mask, the actual smooth and periodic weights, and an additive phase $e(-\alpha z)$ satisfying $N\|\alpha\| > 1$. We write this block without suppressing any coefficient and prove the exact TT^* identity

$$\left| \sum_{z \in I} A(z) e(-\alpha z) \right|^2 = \sum_{|h| < N} e(-\alpha h) \sum_{z, z+h \in I} A(z+h) \overline{A(z)}.$$

The off-diagonal kernel contains four literal Möbius values. Both shifted affine pairs still have determinant h_0 , but no local-support theorem makes this four-sign correlation small.

For a q -invertible phase slope we also prove the complete frequency identity

$$\sum_{r \in \mathbb{F}_q} \left| \sum_{z \in I} A(z) e_q(-r\omega z) \right|^2 = q \sum_{u \in \mathbb{F}_q} \left| \sum_{\substack{z \in I \\ z \equiv u \pmod{q}}} A(z) \right|^2.$$

This is a rigorous average-frequency ceiling, with an explicit aliasing term, but it does not select the literal low-frequency filter or control coherent outer reassembly.

Two obstructions explain why the gate remains open. Genericity alone yields only a constant geometric-series gain and is saturated by bounded phase-matched coefficients. More seriously, a broad prefix-uniform power-saving theorem for all generic twists of $\lambda(n)\lambda(n+h_0)$ would, by Abel summation, imply a power-saving ordinary two-point Chowla estimate. Existing logarithmically averaged or shift-averaged results do not have the literal growing-family quantifiers.

We therefore formulate a restricted hypothesis only for the actual TPC block family and give an exact outer-loss ledger at the numerical exponent $1/200$; the physical TT^* crosswalk remains a separate requirement. The identities, frequency theorem, and obstruction are L0; the literal dictionary is L1. No new L2 fixed- h_0 Möbius estimate is proved, and H3 remains an explicit hypothesis.

Keywords: fixed-shift affine Möbius correlation; generic phase; TT^* ; frequency aliasing; Chowla barrier; parity problem.

Audit convention. L0 means exact finite algebra or a model obstruction, L1 an exact crosswalk to the physical fixed- h_0 carrier, and L2 only a growing arithmetic estimate on that carrier. An average over frequencies, forms, shifts, or scales is not promoted to a result for the prescribed physical coefficient.

1 The H3 target and its literal block

TPC-MVP1 isolates the working target

$$\mathcal{G}_{b,\text{gen}}^{\text{signed}} \ll X Q_b X^{-1/200+o(1)} \quad (1)$$

for every actual long generic packet, together with a separately specified physical TT^* normalization that must yield the 1/400 packet saving [6]. Neither part is known.

TPC-93 supplies the exact generic block. Its resolved key $\xi = (\theta, c, \kappa, r)$ determines an interval $I_\xi \subset \mathbb{Z}$, $N_\xi = |I_\xi|$, and

$$D_\xi(z) = d_\xi + s_\xi z, \quad V_\xi(z) = u_\xi + a_\xi z, \quad (2)$$

$$s_\xi u_\xi - a_\xi d_\xi = h_0 \neq 0, \quad \alpha_\xi = \frac{r\varepsilon_\theta \Omega_\xi}{cq_X} \pmod{1}. \quad (3)$$

The generic condition is

$$N_\xi \|\alpha_\xi\|_{\mathbb{R}/\mathbb{Z}} > 1. \quad (4)$$

The exact coefficient is

$$\begin{aligned} A_{\xi,X}(z) &= \mu(D_\xi(z))\mu(V_\xi(z))\mathbf{1}_{(B_\xi, V_\xi(z))=1} \\ &\quad \times \chi_\theta(\tau_\xi + B_\xi z)W_{\theta,X}(\tau_\xi + B_\xi z), \end{aligned} \quad (5)$$

and the block transform is

$$S_{\xi,X} = \sum_{z \in I_\xi} A_{\xi,X}(z) e(-\alpha_\xi z), \quad e(t) = e^{2\pi i t}. \quad (6)$$

The complete outer coefficient retains

$$\mathbf{c}_{\xi,X} = c_{\theta,X} \mu(\kappa) \mu(B_\xi) \mathbf{m}_{K,X}(r) \zeta_{\xi,X}.$$

The generic low window is exactly

$$\text{Low}_{\text{Gen},X} = \sum_{\xi \in \Xi_{\text{Gen}}} \mathbf{c}_{\xi,X} S_{\xi,X}. \quad (7)$$

These formulas retain both polarizations, all prefixes and masks, the moving coefficients, and one prescribed h_0 [4].

2 Exact TT^* and the four-sign kernel

For $h \in \mathbb{Z}$, define

$$C_{\xi,X}(h) = \sum_{\substack{z \in I_\xi \\ z+h \in I_\xi}} A_{\xi,X}(z+h) \overline{A_{\xi,X}(z)}. \quad (8)$$

Theorem 2.1 (Literal one-block TT^*). *For every resolved block,*

$$\boxed{|S_{\xi,X}|^2 = \sum_{|h| < N_\xi} e(-\alpha_\xi h) C_{\xi,X}(h).} \quad (9)$$

The diagonal is

$$C_{\xi,X}(0) = \sum_{z \in I_\xi} |A_{\xi,X}(z)|^2.$$

For $h \neq 0$, the arithmetic factor in $C_{\xi,X}(h)$ is

$$\mu(D_\xi(z+h))\mu(V_\xi(z+h))\mu(D_\xi(z))\mu(V_\xi(z)), \quad (10)$$

times the two exact coprimality masks, the shifted periodic/local factor

$$\chi_\theta(\tau_\xi + B_\xi(z+h)) \overline{\chi_\theta(\tau_\xi + B_\xi z)},$$

and the corresponding shifted physical weights.

Proof. Expand $|S_{\xi,X}|^2$, set $h = z - z'$, and group terms with the same difference. $\square$

Proposition 2.2 (Fixed determinant survives the shift). *For every integer h , the shifted pair*

$$D_{\xi,h}(z) = D_{\xi}(z + h), \quad V_{\xi,h}(z) = V_{\xi}(z + h)$$

has the same determinant h_0 :

$$s_{\xi}(u_{\xi} + a_{\xi}h) - a_{\xi}(d_{\xi} + s_{\xi}h) = h_0.$$

Proof. The two $a_{\xi}s_{\xi}h$ terms cancel. $\square$

Remark 2.3 (Why TT^* is not yet a saving). TPC-90 classifies the squarefree local support and writes the remaining signs as a Liouville parity correlation [3]. In (10), TT^* raises the correlation order from two signs to four. A useful TT^* argument must gain from frequency or outer averaging; the formal identity alone does not bound the off-diagonal kernel.

3 A complete-frequency theorem and the aliasing ledger

On the nonconstant q_X -invertible branch, literal provenance gives $c \mid \Omega_{\xi}$ and

$$\omega_{\xi} = \varepsilon_{\theta} \frac{\Omega_{\xi}}{c} \pmod{q_X} \in \mathbb{F}_{q_X}^{\times}.$$

Thus the phase is $e_{q_X}(-r\omega_{\xi}z)$.

Theorem 3.1 (Exact full-frequency alias identity). *Let q be prime, $\omega \in \mathbb{F}_q^{\times}$, $I \subset \mathbb{Z}$ finite, and $A : I \rightarrow \mathbb{C}$. Put*

$$S(r) = \sum_{z \in I} A(z) e_q(-r\omega z).$$

Then

$$\sum_{r \in \mathbb{F}_q} |S(r)|^2 = q \sum_{u \in \mathbb{F}_q} \left| \sum_{\substack{z \in I \\ z \equiv u \pmod{q}}} A(z) \right|^2. \quad (11)$$

If the residues of I modulo q are distinct, in particular if I is an interval of length at most q , then

$$\sum_{r \in \mathbb{F}_q} |S(r)|^2 = q \sum_{z \in I} |A(z)|^2. \quad (12)$$

Proof. Expanding the left side and using additive orthogonality gives

$$\sum_{r \in \mathbb{F}_q} e_q(-r\omega(z - z')) = q \mathbf{1}_{\{z \equiv z' \pmod{q}\}},$$

because $\omega \neq 0$. Group the remaining pairs by their residue class. $\square$

Corollary 3.2 (Filtered one-block ceiling). *For any $\mathcal{R} \subseteq \mathbb{F}_q$ and filter m supported on $\mathcal{R}$,*

$$\left| \sum_{r \in \mathcal{R}} m(r) S(r) \right| \leq \|m\|_2 \sqrt{q} \left(\sum_{u \in \mathbb{F}_q} \left| \sum_{\substack{z \in I \\ z \equiv u \pmod{q}}} A(z) \right|^2 \right)^{1/2}. \quad (13)$$

Proof. Apply Cauchy-Schwarz on $\mathcal{R}$, enlarge to all frequencies, and use [theorem 3.1](#). $\square$

Remark 3.3 (Average frequency is not the literal gate). [Equation \(13\)](#) is unconditional, but it has a $\sqrt{q}$ cost and must still be summed with the physical outer coefficients. It does not prove (1); nor may the complete-frequency average be reported as control of one prescribed low-window combination.

4 Two sharp barriers

4.1 Genericity alone gives no power

Proposition 4.1 (Geometric sum and coefficient-blind saturation). *For an interval I of length N ,*

$$\left| \sum_{z \in I} e(-\alpha z) \right| \leq \min \left\{ N, \frac{1}{2\|\alpha\|} \right\}.$$

Hence $N\|\alpha\| > 1$ supplies at most a constant-factor conclusion near the generic boundary. Moreover, among coefficients satisfying $|A(z)| = 1$, the choice $A(z) = e(\alpha z)$ gives

$$\left| \sum_{z \in I} A(z) e(-\alpha z) \right| = N.$$

Proof. The first statement is the standard finite geometric-series bound. The second is immediate term by term. $\square$

The saturation coefficient is not claimed to be the literal Möbius coefficient. It proves that generic phase, bounded weights, and interval length alone cannot imply a power saving.

4.2 A broad generic theorem reaches ordinary Chowla

Lemma 4.2 (Twisted-prefix removal). *Let $|b_n| \leq 1$, $0 < \delta < 1$, and $\alpha_N = N^{-1+\delta}$. If*

$$\max_{1 \leq T \leq N} \left| \sum_{n \leq T} b_n e(\alpha_N n) \right| \leq C N^{1-\eta},$$

then

$$\left| \sum_{n \leq N} b_n \right| \leq C'(1 + N^\delta) N^{1-\eta} \ll N^{1-\eta+\delta}. \quad (14)$$

Proof. Let $B(T) = \sum_{n \leq T} b_n e(\alpha_N n)$. Partial summation applied to $b_n = B(n) - B(n-1)$ after removal of the twist gives

$$\left| \sum_{n \leq N} b_n \right| \leq \max_{T \leq N} |B(T)| (1 + N|1 - e(-\alpha_N)|).$$

Use $|1 - e(-t)| \leq 2\pi\|t\|$. $\square$

Corollary 4.3 (Broad-family Chowla barrier). *Let λ denote the Liouville function. Fix $h_0 > 0$. Suppose a theorem gave, for some $\eta > 0$, the preceding prefix bound with*

$$b_n = \lambda(n)\lambda(n + h_0)$$

for every sufficiently large N and every generic phase $\alpha_N = N^{-1+\delta}$, where $0 < \delta < \eta$. Then

$$\sum_{n \leq N} \lambda(n)\lambda(n + h_0) \ll N^{1-(\eta-\delta)},$$

a power-saving ordinary two-point Chowla estimate.

Proof. Since $N\|\alpha_N\| = N^\delta > 1$, these twists are generic. Apply [lemma 4.2](#). $\square$

Remark 4.4 (Restricted does not mean equivalent). The literal H3 family contains only the phases, forms, masks, and weights generated by the TPC dictionary. It is not asserted to contain every pair $(n, n + h_0)$ and every $\alpha_N = N^{-1+\delta}$. Therefore [corollary 4.3](#) does not prove H3 equivalent to Chowla. It explains why a convenient uniform theorem over arbitrary affine forms and generic phases would be unexpectedly deep.

5 Why existing averaged theorems do not fill the gap

The logarithmically averaged two-point Chowla theorem controls fixed affine forms with reciprocal weight [2]. Shift-averaged results average the prescribed displacement [1]. The literal TPC blocks instead require:

- (i) growing slopes and intercepts but one prescribed nonzero determinant h_0 ;
- (ii) ordinary smooth physical weights and actual interval origins;
- (iii) uniform control of every prefix needed by partial summation;
- (iv) exact content and coprimality masks;
- (v) the specified low-frequency multiplier and both polarizations; and
- (vi) coherent outer reassembly at a quantitative power rate.

TPC-96 proves that these quantifiers cannot be supplied by an anchor-free Tauberian conversion [5]. This comparison is a quantifier audit, not a criticism of the averaged theorems.

6 The restricted hypothesis and an exact loss ledger

For a block, put

$$V_{\xi,X} = \sum_{z \in I_\xi} |A_{\xi,X}(z)|, \quad \Delta_{\xi,X} = \begin{cases} |S_{\xi,X}|/V_{\xi,X}, & V_{\xi,X} > 0, \\ 0, & V_{\xi,X} = 0. \end{cases}$$

Define the actual outer absolute mass

$$\mathcal{M}_{\text{Gen},X} = \sum_{\xi \in \Xi_{\text{Gen}}} |\mathfrak{c}_{\xi,X}| V_{\xi,X}. \quad (15)$$

Theorem 6.1 (Exact termwise exponent ledger). *If, for fixed exponents $\eta, \ell_{\text{out}} \geq 0$,*

$$\sup_{\xi \in \Xi_{\text{Gen}}} \Delta_{\xi,X} \leq X^{-\eta+o(1)}, \quad (16)$$

$$\mathcal{M}_{\text{Gen},X} \leq X^{\ell_{\text{out}}+o(1)} X Q_X, \quad (17)$$

then

$$|\text{Low}_{\text{Gen},X}| \leq X Q_X X^{-(\eta-\ell_{\text{out}})+o(1)}. \quad (18)$$

Thus this termwise route reaches the numerical exponent $1/200$ for the displayed linear generic window if $\eta - \ell_{\text{out}} \geq 1/200$. Identifying that bound with the normalized quadratic object in (1) still requires the exact physical TT^* crosswalk and prefactor.

Proof. Apply the triangle inequality to (7), insert $|S_{\xi,X}| = \Delta_{\xi,X} V_{\xi,X}$, and use (16)–(17). $\square$

Remark 6.2 (Sufficient, deliberately not necessary). **Theorem 6.1** discards cancellation between blocks. A coherent TT^* theorem may be strictly stronger. Conversely, the scale $X Q_X$ in (17) is not asserted for the physical coefficient; it is a separately auditable normalization condition.

Working hypothesis 6.3 (Restricted actual-family arithmetic gate). For every actual generic packet b , the complete normalized signed TT^* object built from (5)–(7), including all prefixes, satisfies

$$\mathcal{G}_{b,\text{gen}}^{\text{signed}} \ll X Q_b X^{-1/200+o(1)}.$$

The exact physical TT^* prefactor is recorded separately and converts this quadratic saving to the packet target without an unrecorded loss.

Remark 6.4. **Working hypothesis 6.3** is H3 stated on the now explicit block dictionary. It is not proved in this paper. The point is to prevent its replacement by a theorem for arbitrary weights, fixed forms, averaged shifts, or a complete-frequency mean.

7 Deterministic regression certificate

The script `experiments/tpc108_certificate.py` verifies:

- the direct and shifted-correlation forms of [theorem 2.1](#);
- the full-frequency alias identity, including an interval longer than q ;
- coefficient-blind saturation in [proposition 4.1](#); and
- the finite Abel inequality underlying [lemma 4.2](#).

The output is an algebraic regression certificate only.

8 Claim boundary and next gate

Proved

The literal generic block has an exact TT^* correlation kernel with four Möbius factors, and both shifted pairs preserve determinant h_0 . Complete-frequency averaging has the exact alias formula (11). Genericity alone cannot give a power saving, and a broad prefix-uniform generic-twist theorem would imply a power-saving ordinary Chowla estimate. The termwise outer-loss law (18) is exact. These results are L0; their attachment to the imported TPC-93 coefficient is L1.

Not proved

This paper does not prove:

- [working hypothesis 6.3](#) or (1);
- a growing fixed- h_0 Möbius or Liouville cancellation estimate;
- the physical outer-mass bound (17);
- conversion of a complete-frequency mean into the literal low-window combination;
- the separate 1/400 packet normalization, full-block return, determinant/zero-mode compatibility, or endpoint budget; or
- a parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem.

Nothing is specialized to $h_0 = 2$.

Verdict

TPC-108 advances the route by replacing the phrase “generic affine dispersion” with one exact arithmetic object and two precise obstructions. It does not open H3. The next rational move is TPC-109’s coherent prime-square block test: exploit special structure of one actual residual block, or prove a coefficient-blind saturation theorem. Such a theorem stops only the subroute based on support, moduli, occupancy, and diagonal energy; it does not rule out gains from literal arithmetic phase relations or a proved dominant-prime statistic.

References

- [1] Kaisa Matomäki, Maksym Radziwiłł, and Terence Tao. An Averaged Form of Chowla’s Conjecture. *Algebra & Number Theory*, 9(9):2167–2196, 2015. doi: 10.2140/ant.2015.9.2167. URL <https://doi.org/10.2140/ant.2015.9.2167>.
- [2] Terence Tao. The Logarithmically Averaged Chowla and Elliott Conjectures for Two-Point Correlations. *Forum of Mathematics, Pi*, 4:e8, 2016. doi: 10.1017/fmp.2016.6. URL <https://doi.org/10.1017/fmp.2016.6>.
- [3] Liang Wang. Primitive Fixed-Determinant Affine Möbius Pairs: Exact Local Obstructions, Squarefree Support, and the Remaining Parity Correlation, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-90-primitive-affine-local-obstruction-split. Preprint.
- [4] Liang Wang. Literal Low-Window Affine Export: Content-Factor Extraction, Resonance Sectors, and Fixed-Determinant Preservation, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-93-literal-low-window-affine-export. Preprint.
- [5] Liang Wang. Anchored Logarithmic Primitives on Translated Fibers: Exact Tauberian Defects, Uniformity Regimes, and the Literal Reassembly Gate, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-96-anchored-tauberian-translation-defect. Preprint.
- [6] Liang Wang. TPC-MVP1: A Falsifiable Proof Architecture for Fixed-Shift Prime-Pair Correlations, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-mvp1-fixed-shift-proof-architecture. Preprint.